The general structure:

$$T = \begin{bmatrix} -4 & 1 & & & \\ 1 & -4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & -4 \end{bmatrix} \in \mathbb{R}^{(N-1)\times(N-1)}, \quad L_h = -\frac{1}{h^2} \begin{bmatrix} T & I & & & \\ I & T & I & & \\ & I & \ddots & \ddots & \\ & & \ddots & \ddots & I \\ & & & I & T \end{bmatrix} \in \mathbb{R}^{(N-1)^2\times(N-1)^2}$$

$$F = [f_{11}, \dots, f_{N-1,1}, f_{12}, \dots, f_{N-1,2}, \dots, f_{N-1,N-1}]^T + \frac{1}{h^2}[g_{01}+g_{10}, g_{20}, \dots, g_{N-1,0}+g_{N,1}, \dots]^T \in \mathbb{R}^{(N-1)^2}$$

The matrix A can also be compactly written via the Kronecker product $\otimes$

$$C \otimes D := \begin{bmatrix} c_{11} D & \cdots & c_{1m} D \\ c_{21} D & & \vdots \\ \vdots & & \vdots \\ c_{n1} D & \cdots & c_{nm} D \end{bmatrix}$$

as follows: $L_h = -\frac{1}{h^2}(I_{N-1} \otimes S + S \otimes I_{N-1})$, where $S = \begin{bmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & -2 \end{bmatrix} \in \mathbb{R}^{(N-1)\times(N-1)}$

Advantages:
- reduction of storage since only matrices of dimension $(N-1)\times(N-1)$ are needed
- allows for d-dimensional generalization, e.g., 3D-Laplace via

$$I_{N-1} \otimes I_{N-1} \otimes S + I_{N-1} \otimes S \otimes I_{N-1} + S \otimes I_{N-1} \otimes I_{N-1}$$

Lecture 08

06.11.2025

Convergence analysis for finite difference schemes

Goal: we want that approximate solution u_h is "close" to exact solution u!

Questions:
- how do we define "close"?
- how do we compare $u_h : \overline{\Omega}_h \to \mathbb{R}$ with $u : \overline{\Omega} \to \mathbb{R}$?

Notation:
- in the following, we sometimes use N and h to denote the dimensions of a matrix and a (small) grid parameter $h > 0$ without necessarily restricting ourselves to a specific (one-dimensional) grid configuration

For comparison of the discrete approximation defined only on the grid and the exact solution being a continuous function, we introduce a restriction operator

$$R_h : C(\overline{\Omega}) \to \mathbb{R}^{N-1}$$

which in the spatially one-dimensional case is defined by

$$R_h w = \begin{bmatrix} w(a+h) \\ w(a+2h) \\ \vdots \\ w(b-h) \end{bmatrix} \quad \text{or} \quad R_h w = \begin{bmatrix} w(a) \\ w(a+h) \\ \vdots \\ w(b) \end{bmatrix}$$

How do we then "compare" the two vectors u_h and $R_h u$?

<u>Definition</u> (norm on a vector space) A <u>norm</u> on an $\mathbb{R}$- or $\mathbb{C}$ vector space V is a mapping $\|\cdot\| : V \to \mathbb{R}$ s.t.

a) $\|v\| \geq 0 \quad \forall v \in V$ and $\|v\| = 0 \Leftrightarrow v = 0$.

b) $\|\alpha v\| = |\alpha| \, \|v\|$ for all $\alpha \in \mathbb{R}$ (or $\mathbb{C}$) and $v \in V$.

c) $\|v+w\| \leq \|v\| + \|w\|$ for all $v, w \in V$ ("triangle inequality").

Given a norm $\|\cdot\|$ on $\mathbb{R}^d$, we can define an <u>operator/matrix norm</u> on $\mathbb{R}^{d \times d}$ by

$$|||A||| := \sup_{\substack{v \in \mathbb{R}^d \\ \|v\|=1}} \|Av\| = \sup_{\substack{v \in \mathbb{R}^d \\ v \neq 0}} \frac{\|Av\|}{\|v\|}$$

generally different norms!

For a matrix norm $|||\cdot|||$ induced by vector norms $\|\cdot\|$ it holds

$$|||A \cdot B||| \leq |||A||| \cdot |||B||| \quad \text{and} \quad \|Av\| \leq |||A||| \cdot \|v\| .$$

Given some family/sequence $h > 0$, e.g., $h = \frac{1}{N}$ for varying $N \in \mathbb{N}$, we distinguish a single norm $\|\cdot\|_h$ on $\mathbb{R}^N$ and a <u>system of norms</u> $(\|\cdot\|_h)_{h>0}$.

<u>Example</u>
- the <u>discrete L^2-norm</u> on $\mathbb{R}^N$: $\|v\|_{2,h} = \left(h \sum_{i=1}^{N} |v_i|^2\right)^{1/2}$
- the <u>maximum norm</u> on $\mathbb{R}^N$: $\|v\|_{\infty,h} = \max_{1 \leq i \leq N} |v_i|$, $v = \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix}$.

<u>Definition</u> Let $(\|\cdot\|_h)_{h>0}$ be a system of norms. A finite difference method $L_h u_h = f_h$ approximating $Lu = f$ with exact solution u is called

- <u>consistent</u> w.r.t. $(\|\cdot\|_h)_{h>0}$, if $\|f_h - L_h R_h u\|_h \to 0$ as $h \to 0$,
- <u>consistent of order $p>0$</u> w.r.t. $(\|\cdot\|_h)_{h>0}$, if $\|f_h - L_h R_h u\|_h \in \mathcal{O}(h^p)$ as $h \to 0$,
- <u>convergent</u> w.r.t. $(\|\cdot\|_h)_{h>0}$, if $\|u_h - R_h u\|_h \to 0$ as $h \to 0$,
- <u>convergent of order $p>0$</u> w.r.t. $(\|\cdot\|_h)_{h>0}$, if $\|u_h - R_h u\|_h \in \mathcal{O}(h^p)$ as $h \to 0$,
- <u>stable</u> w.r.t. $(\|\cdot\|_h)_{h>0}$ if $L_h \in \mathbb{R}^{(N-1)\times(N-1)}$ is invertible and if there exists $c > 0$ with $|||L_h^{-1}|||_h \leq c < \infty$ for <u>eventually</u> all $h > 0$.

  ↳ independent of h!

Example consider $\underbrace{-u''}_{Lu} = f$, $u(a) = 0 = u(b)$

Here, $L_h u(x) = -\dfrac{u(x+h) - 2u(x) - u(x-h)}{h^2}$ so that $L_h = \frac{1}{h^2}\begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{pmatrix}$, $u_h = \begin{pmatrix} u(a+h) \\ \vdots \\ u(b-h) \end{pmatrix}$

and $f_h = (f(a+h), \dots, f(b-h))^T$.

Consistency? We have to check $L_h R_h u - f_h$ which is here given as

$$L_h R_h u - f_h = L_h R_h u - R_h f = L_h R_h u - R_h \overset{u \text{ is exact solution}}{L u} = L_h R_h u - R_h(-\Delta u)$$

$$= \frac{1}{h^2}\begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{pmatrix}\begin{pmatrix} u(a+h) \\ \vdots \\ u(b-h) \end{pmatrix} - \begin{pmatrix} u''(a+h) \\ \vdots \\ u''(b-h) \end{pmatrix} \rightsquigarrow \text{Taylor expansion } \mathcal{O}(h^2)$$

$$\Rightarrow \quad \| L_h R_h u - f_h \|_{\infty,h} = \max_{1 \le i \le N-1} \left\| u''(a+ih) - \frac{1}{h^2}\big(u(a+(i+1)h) - 2u(a+ih) + u(a+(i-1)h)\big) \right\|$$

$\in \mathcal{O}(h^2)$ $\rightarrow$ consistent of order 2 w.r.t. the <u>maximum norm</u>! (if $u \in C^4$!)